

Silpa Silpa

SQL Aggregate Function and Operators

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Data Analyst

Q1 Student Table

Students

Student-id	name	Age	Department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q

SELECT DISTINCT
FROM Students;

Department
IT
HR
Finance

Q SELECT department,
Avg (Age) AS avg-age
FROM Students;
GROUP BY department;

Department	avg-age
IT	20.5
HR	22
Finance	23

(05) SELECT department,
 Count(*) as Student_Count
 FROM Students
 GROUP BY department
 Having Student_Count > 1 ;

department	Student_Count
IT	2
HR	2

④ SELECT *
 FROM Students
 WHERE age BETWEEN 21 AND 23 ;

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ SELECT *
 FROM Students
 WHERE department = 'IT' OR department = 'HR' AND age > 21 ;

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

⑥ The Course table

Courses			
Course_id	Course_name	Department	Credits
101	SQL Basic	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

⑥ SELECT department,
Sum (Credits) AS total_credits

FROM Courses

GROUP BY department

HAVING total_credits > 5;

Department	total_credits
IT	11

⑦ SELECT *

FROM ~~select~~ Courses

WHERE Credits NOT (4);

Course_id	Course_name	Department	Credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑩ The Enrollments Table

⑪ SELECT MAX (grade) AS Max-grade,
MIN (grade) AS Min-grade,
AVG (grade) AS Avg-grade
FROM enrollments ;

Max-grade	Min-grade	Avg-grade
90	75	84.6

⑫ SELECT Course_id,
COUNT (enrollments) AS enrollment-count
FROM enrollments
GROUP BY Course_id ;

Course_id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Q2) The Salaries Table Part

(11) SELECT department,
Sum (Salary) as total-Salary
Sum (Bonus) as total-Bonus

FROM Salaries

GROUP BY department ;

department	total-Salary	total-Bonus
IT	122 000	11 000
HR	109 000	7 500
Finance	70 000	6 000

(12) SELECT department,
Avg (Salary) as avg-Salary

FROM Salaries

GROUP BY department

HAVING avg-Salary > 55 000

department	avg-Salary
IT	61 000
HR	54 500
Finance	70 000

(13) SELECT employee_id,
Name,
Salary,
bonus,
Salary + Bonus as total-Compensation

FROM Salaries

~~GROUP BY~~

HAVING total-Compensation

Employeeid	name	Salary	bonus	total Compensation
1	Tom	65000	5000	65000
3	Spide	70000	6000	76000
4	Tyke	62000	5500	67500

as the Project table Part

(14) **SELECT** department,
 Sum(budget) AS total-budget-1
 AVG(budget) AS ~~total-bud~~ avg-budget-1
FROM Projects
Group By department
 Having avg-budget > 70000 ;

department	total - budget	avg - budget
IT	270000	135000
Finance	80000	80000

(15) **SELECT** *
FROM Projects
WHERE budget BETWEEN 80000 AND 120000
 department != 'Marketing' ;

Projectid	ProjectName	department	budget
1	AI App	IT	120000
2	Payrol System	Finance	80000
3	Dashboard	IT	150000
5	HR Portal	HRC	80000